Repository

eason: 4 · Enterprise · Category: Enterprise · Level: Intermediate · Build with Waqa

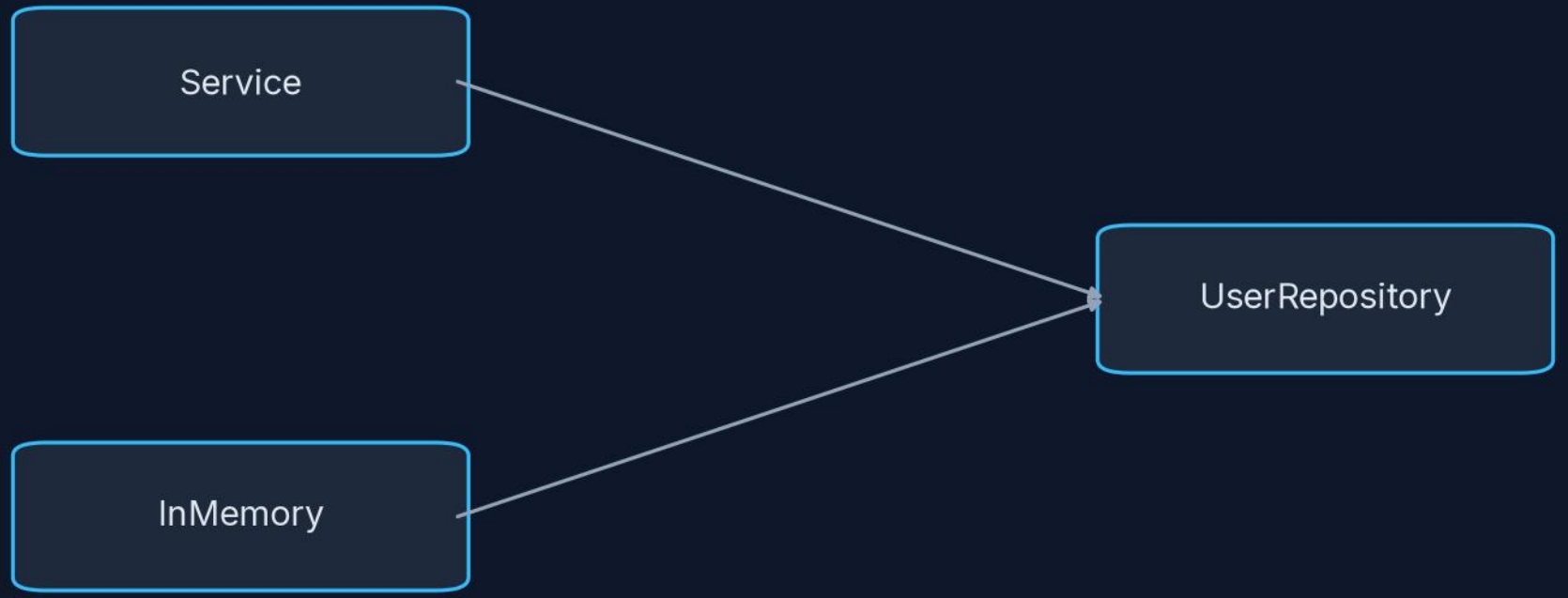
Today's Hook

- Today: Repository as a StreamForge product force — not trivia.
- Runnable Demo on JDK 25 with captured stdout.
- Classic encoding → Java 25 / Zazr modernization → trade-offs.
- CapCut slides + full spoken script included.

The Problem

- Naive local fixes: more ifs, statics, nulls, or threads.
- Illegal states still compile; tests become order-dependent.
- Design reviews lack a shared name for the force.
- Interviews punish memorization without judgment.

Structure



Repository structure

Real-World Scenario

- StreamForge creator platform is the running metaphor.
- Same names appear in naive code, refactor, and Demo output.
- Stay in one story for the full 10–12 minutes.
- Product language first; pattern vocabulary second.

Naive Code

```
// Naive sketch for Repository
public class Naive {
  // duplicated logic / leaked details / unsafe sharing
  Object doIt(Object... args) {
    if (args == null) return null;
    return args[0];
  }
}
```

Why Naive Hurts

- Coupling spreads across controllers and infrastructure.
- Tests need sleeps or static reset hooks.
- New variants require shotgun edits.
- On-call cannot reason about failure modes.

Implementation

```java
// file: src/main/java/com/buildwithwaqar/patterns/enterprise/repository/InMemoryUserRepository.java
package com.buildwithwaqar.patterns.enterprise.repository;

import java.util.Collection;
import java.util.Map;
import java.util.Optional;
import java.util.concurrent.ConcurrentHashMap;

record User(String id, String email) {}

interface UserRepository {
    Optional<User> findById(String id);
    User save(User user);
    Collection<User> findAll();
}

/** Collection-like interface over persistence details. */
public final class InMemoryUserRepository implements UserRepository {
    private final Map<String, User> store = new ConcurrentHashMap<>();

    public Optional<User> findById(String id) { return Optional.ofNullable(store.get(id)); }
    public User save(User user) { store.put(user.id(), user); return user; }
    public Collection<User> findAll() { return store.values(); }
}
```

Demo.main

```
package com.buildwithwaqar.patterns.enterprise.repository;
public final class Demo {
  public static void main(String[] a) {
    System.out.println("=== Repository walkthrough ===");
    var repo = new InMemoryUserRepository();
    repo.save(new User("u1", "waqar@bwq.dev"));
    repo.save(new User("u2", "editor@bwq.dev"));
    System.out.println("find=" + repo.findById("u1").orElseThrow());
    System.out.println("missing=" + repo.findById("nope"));
    System.out.println("count=" + repo.findAll().size());
  }
}
```

Walkthrough Output

```
=== Repository walkthrough ===
find=User[id=u1, email=waqar@bwq.dev]
missing=Optional.empty
count=2
```

Step-by-Step Trace

- Each stdout line maps to a refactor stage.
- Re-run Demo locally on JDK 25 to verify.
- Differences usually mean a dirty tree — pull main.
- Pause the video and predict the next line before revealing.

Modern Java 25 / Zazr

```
// Java 25 / Zazr direction for Repository
// records + sealed + switch
// Option/Either/Validation instead of null
// virtual threads for blocking fan-out
// inject interfaces at the edge
```

Trade-offs

- Clarity vs ceremony — do not spawn five types for two lines.
- Closed sealed worlds vs open SPI plugins.
- Teaching demos omit production idempotency/metrics.
- Prefer simpler language features when they erase the force.

Related Patterns

- Contrast with the neighbor pattern (force difference).
- Name the roles from memory in sixty seconds.
- Point to the season map for what comes next.
- Link the GitHub episode folder in the description.

Recap

- Force → naive pain → Repository → Demo stdout.
- Modernize with Java 25 / Zazr when it shrinks code.
- Teach when NOT to use it.
- Practice: break a test, fix it, re-run Demo.